

Scalability & Future Plan

Date	05 July 2026
Team ID	SWTID-2026-2423
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Centralized Database Storage	Storing verified credentials on a standard central server introduces a single point of failure and potential data tampering.	High
2	Manual Institute Onboarding	Universities and companies must be manually registered by an admin, slowing down platform adoption.	Medium
3	No Offline Verification	Verification of a user's wallet badges strictly requires an active, stable internet connection to fetch records.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-instance server handling limited concurrent login requests from one institution.	Deploy on cloud infrastructure (e.g., AWS/Azure) with Auto-Scaling Groups and a Load Balancer to manage thousands of active students and verifiers.
2	Data Storage	Local database storing all credential metadata, PDF certificates, and profile images.	Migrate file storage to a decentralized network like IPFS (InterPlanetary File System) or cloud object storage (AWS S3) to preserve bandwidth and storage space.
3	Performance	Direct database queries for every verification check, which can lag under high query volumes.	Implement Redis Caching for frequently accessed public profiles and pre-fetch digital wallet verification states.
4	Security	Standard encrypted passwords and simple role-based API access.	Integrate Blockchain-based smart contracts (e.g., Polygon/Ethereum) to issue unalterable cryptographic hashes of the certificates, ensuring complete tamper-proofing.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Decentralized Identifiers (DIDs) & Blockchain Hash Vaulting	Next 2-3 Months	Total data immutability. No one—not even SkillWallet admins—can fake or alter a badge once issued.
Phase 3	Automated API Integrations for Universities	Next 6 Months	Seamless linking with LMS platforms (like Moodle, Canvas, or Coursera) to auto-issue SkillWallet badges upon course completion.
Phase 4	AI Career & Skill Path Recommendation Engine	Next 9-12 Months	Analyzes a student's current SkillWallet badges to recommend specific industry courses and job openings that match their verified skills.